

Data Mining & Business Intelligence (IT647)
Selected Topics In Data Science (IT688)
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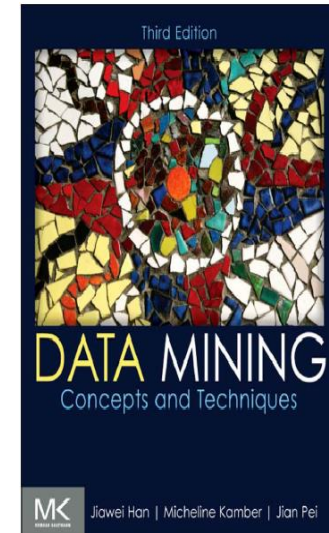
Data Mining & Business Intelligence Course_IT674

Data Mining

— Chapter 10 —

Cluster Analysis: Basic Concepts and Methods

Text Book: “Data Mining: Concepts and Techniques”, Third Edition



Outlines

- Clustering: Basic Concepts 
- Basic Clustering Methods
- K-means Clustering
- Summary

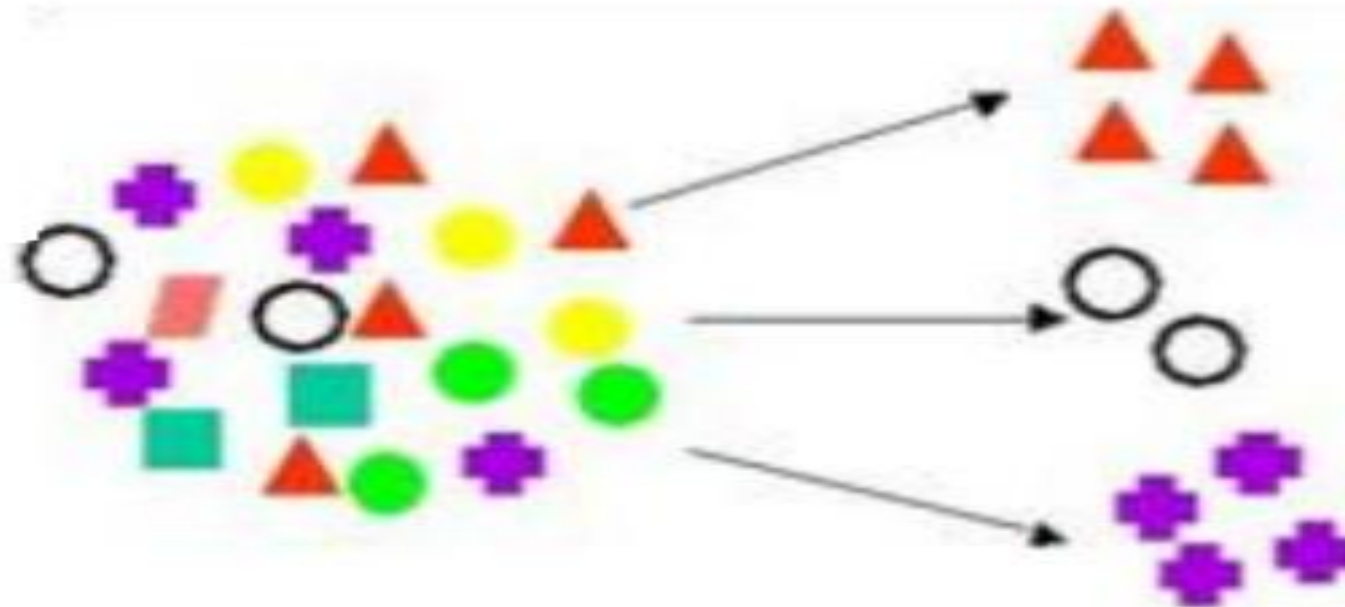
Supervised vs. Unsupervised Learning

- **Supervised learning (classification)**
 - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
 - New data is classified based on the training set
- **Unsupervised learning (clustering)**
 - The class labels of training data is **unknown**
 - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

What is clustering?

- **Clustering:** the process of grouping a set of objects into classes of similar objects
 - Documents within a cluster should be similar.
 - Documents from different clusters should be dissimilar.
- The commonest form of *unsupervised learning*
 - Unsupervised learning = learning from raw data, as opposed to supervised data where a classification of examples is given
- A common and important task that finds many applications in IR and other places

Examples of Clustering



Quality: What is Good clustering?

- A good Clustering method will produce high quality clusters
 - High intra-class similarity: cohesive within clusters
 - Low inter-class similarity: distinctive between clusters
- The quality of a clustering method depends on
 - The similarity measure used by the method
 - Its implementation,
 - Its ability to discover some or all of the hidden patterns

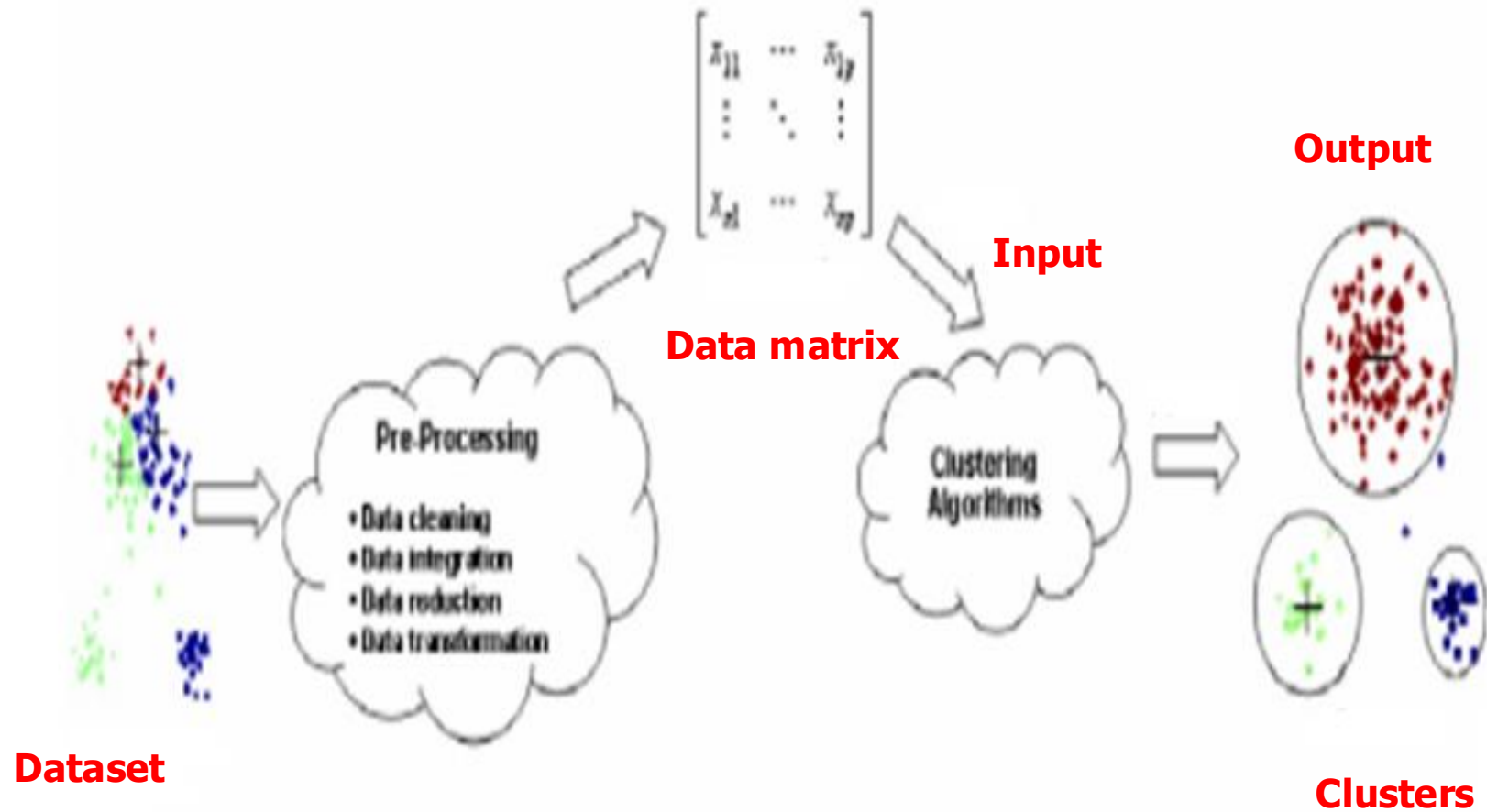
Data types in Cluster Analysis

- Data matrix (or object-by- variable structure)
- Interval- Scaled variables (tempereture)
- Binary variables (0,1)
- A categorical variable(red , green ,blue)
- A discrete ordinal variable (small, medium, large)
- A ration-scaled variable (weight , age)

Applications

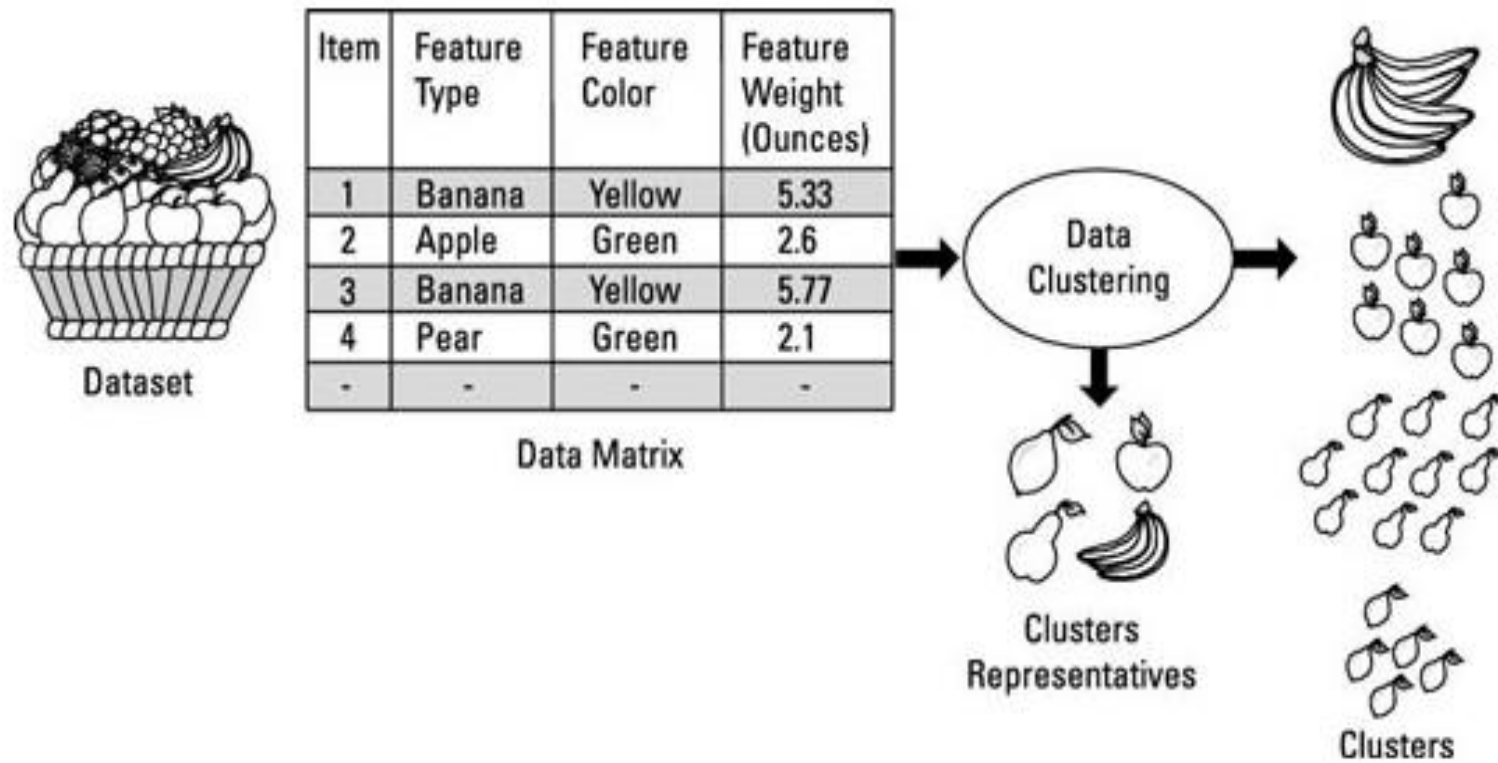
- Information Retrieval (documents classification,...)
- Pattern Recognition
- Spatial Data Analysis
- Image Processing
- Economic Science (especially market research)
- BioInformatics
- Medical Imaging
-

Clustering process



Clustering process

Example



Requirements for Cluster Analysis

Typical **requirements** of clustering in data mining:

- Clustering Scalability;
- Capability of clustering high-dimensionality data;
- Ability to deal with different types of attributes;
- Ability to deal with unstructured and noisy data;
- Interpretability and usability.

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Basic Clustering Methods

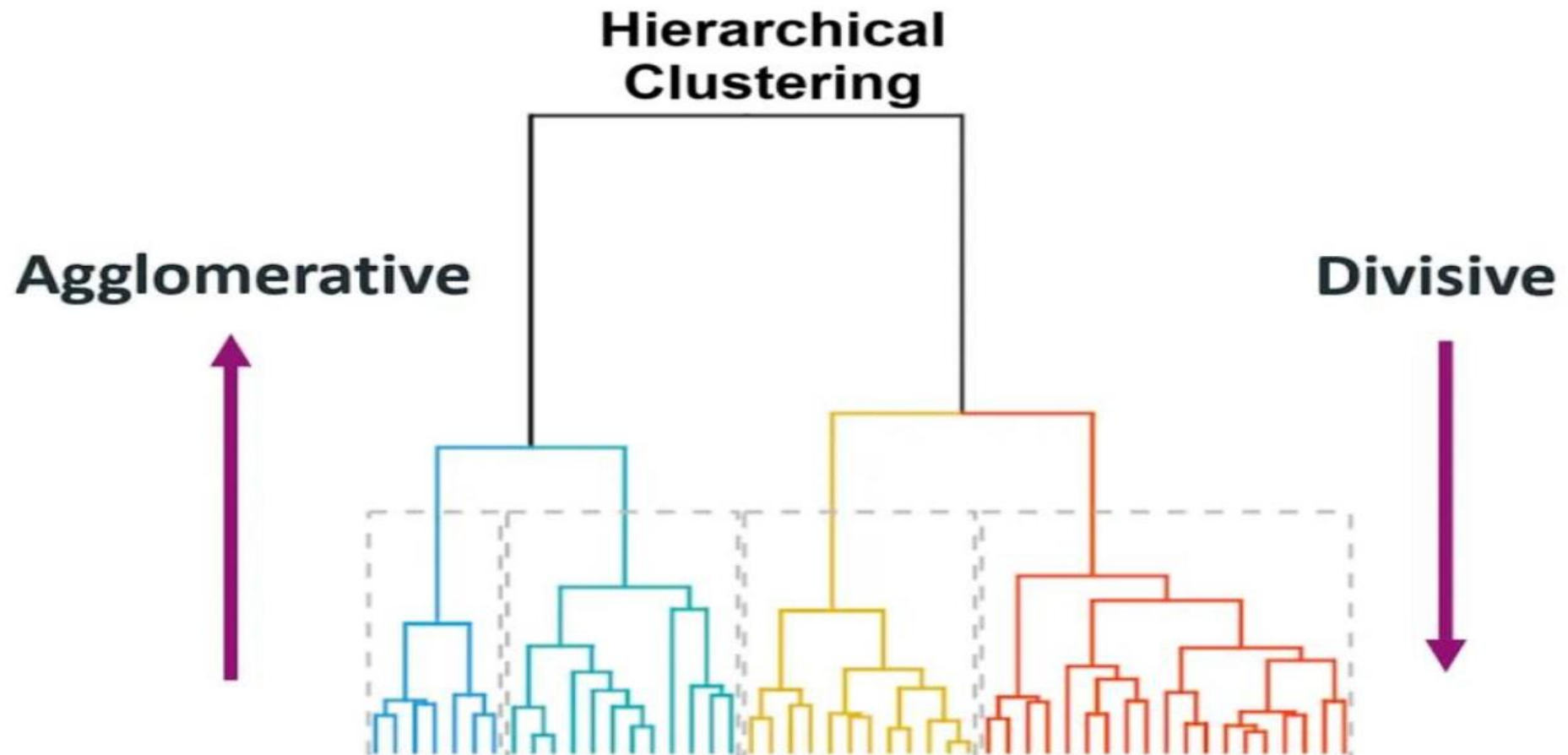
- **Partitioning approach**
 - given a set of n objects, a partitioning method constructs k partitions of the data, where each partition represents a cluster and $k \leq n$.
 - divides the data into k groups such that each group must contain at least one object.
 - adopt exclusive cluster separation; each object must belong to exactly one group.
 - most partitioning methods are distance-based.
 - ***Typical methods:*** K-means, K-medoids, CLARANS

Basic Clustering Methods

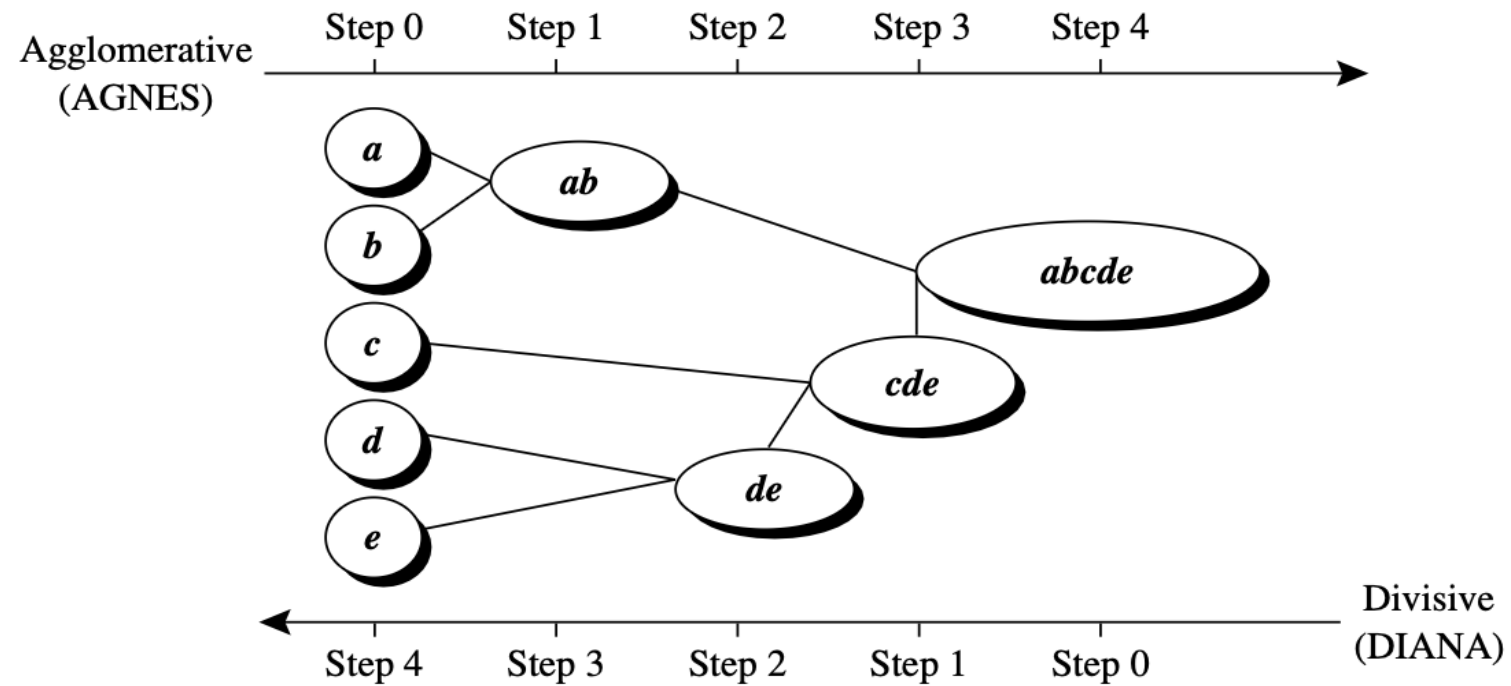
- **Hierarchical approach**

- creates a hierarchical decomposition of the given set of data objects.
- A hierarchical method can be classified as being either ***agglomerative*** or ***divisive***, based on how the hierarchical decomposition is formed.
- agglomerative approach: (bottom-up approach)
 - starts with each object forming a separate group;
 - merges the objects or groups close to one another, until all the groups are merged into one or a termination condition holds.
- divisive approach: (top-down approach)
 - starts with all the objects in the same cluster.
 - in each successive iteration, a cluster is split into smaller clusters, until eventually each object is in one cluster, or a termination condition holds.

Basic Clustering Methods



Basic Clustering Methods



Basic Clustering Methods

- **Hierarchical approach**

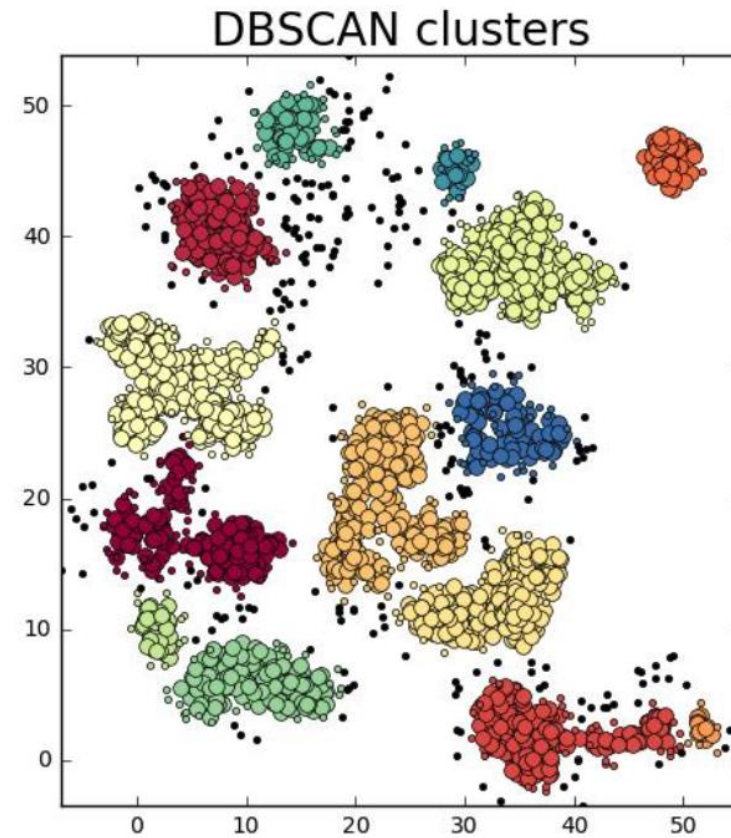
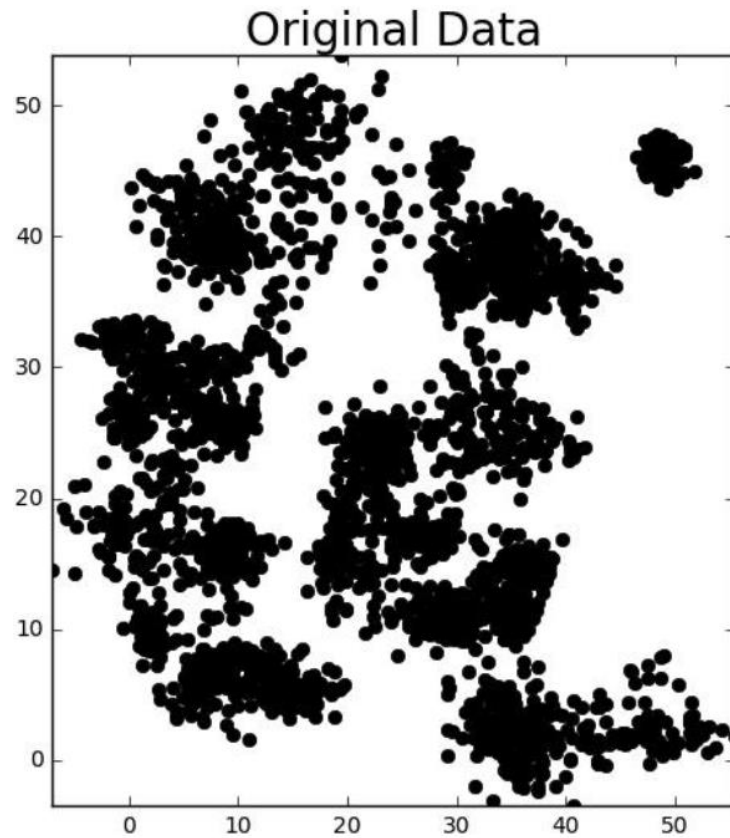
- Hierarchical methods suffer from the fact that once a step (merge or split) is done, it can never be undone.
- Such Hierarchical techniques cannot correct erroneous decisions
- *Typical methods*: Diana, Agnes, BIRCH, CAMELEON

Basic Clustering Methods

- **Density based approach**

- based on connectivity and density functions
- divide a set of objects into multiple exclusive clusters, or a hierarchy of clusters.
- can find arbitrarily shaped clusters
- Clusters are dense regions of objects in space that are separated by low-density regions
- Cluster density: Each point must have a minimum number of points within its “neighborhood”
- ***Typical methods:*** DBSACN, OPTICS, DenClue

Basic Clustering Methods



Basic Clustering Methods

- **Model based**

- a model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
- Typical methods: EM, SOM, COBWEB

- **Frequent pattern-based**

- Based on the analysis of frequent patterns
- Typical methods: p-Cluster

- **Link-based clustering**

- Objects are often linked together in various ways
- Massive links can be used to cluster objects: SimRank, LinkClus.

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K-Means: A Centroid-Based Technique

- used for clustering **numerical data**, usually a set of measurements about objects of interest.
- **Input:** numerical. There must be a distance metric defined over the variable space.
 - Euclidian distance
- **Output:** The centers of each discovered cluster, and the assignment of each input datum to a cluster.
 - Centroid

K-Means: A Centroid-Based Technique

- In K-means we define two measures of distances, between two data points(records) and the distance between two clusters.
- Distance can be measured (calculated) in a number of ways but four principles tend to hold true:
 1. Distance is not negative (it is stated as an absolute value)
 2. Distance from one record to itself is zero.
 3. Distance from record I to record J is the same as the distance from record J to record I, again since the distance is stated as an absolute value, the starting and end points can be reversed.
 4. Distance between two records can not be greater than the sum of the distance between each record and a third record.
- Euclidean distance: is the most popular method for calculating distance. Euclidian distance is a “ordinary” distance that one could measure with a ruler.

K-Means: A Centroid-Based Technique

- suppose a data set, D , contains n objects in Euclidean space.
- distributes the objects in D into k clusters, $C_1, \dots, C_k$, that is, $C_i \subset D$ and $C_i \cap C_j = \emptyset$; for $(1 \leq i, j \leq k)$.
- an objective function is used to assess the partitioning quality so that objects within a cluster are similar to one another but dissimilar to objects in other clusters.
 - aims for high intracluster similarity and low intercluster similarity.

K-Means: A Centroid-Based Technique

- K-Means uses the centroid of a cluster, C_i , to represent that cluster,
 - the centroid of a cluster is its center point.
- the difference between an object $p \in C_i$ and c_i (representative of the cluster) is measured by $\text{dist}(p, c_i)$, where $\text{dist}(x,y)$ is the Euclidean distance between two points x and y .
- the quality of cluster C_i can be measured by the within-cluster variation, which is the sum of squared error between all objects in C_i and the centroid c_i , defined as:

$$E = \sum_{i=1}^k \sum_{p \in C_i} \text{dist}(p, c_i)^2$$

- E is the sum of the squared error for all objects in the data set,
- p is the point in space representing a given object,
- c_i is the centroid of cluster C_i

How does the k-means algorithm **work**?

- The k-means algorithm defines the centroid of a cluster as the mean value of the points within the cluster:
 1. first, it randomly selects k of the objects in D , each of which initially represents a cluster mean or center,
 2. for each of the remaining objects, an object is assigned to the cluster to which it is the most similar, based on the Euclidean distance between the object and the cluster mean,
 3. for each cluster, it computes the new mean using the objects assigned to the cluster in the previous iteration.
 4. all the objects are then reassigned using the updated means as the new cluster centers.
 5. the iterations continue until the assignment is stable (the clusters formed in the current round are the same as those formed in the previous round).

Algorithm: K-means

Input:

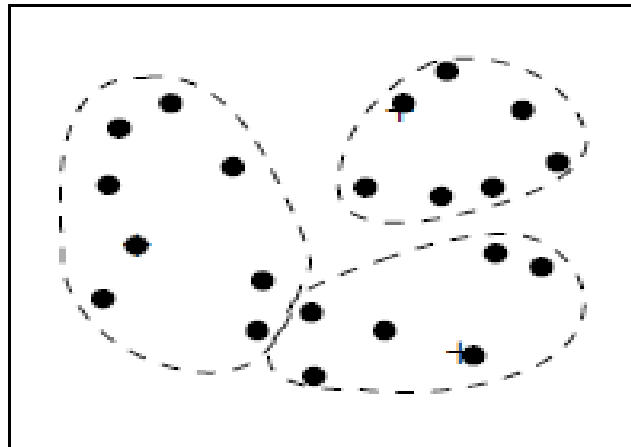
- k: the number of clusters,
- D: a data set containing n objects.

Output: A set of k clusters

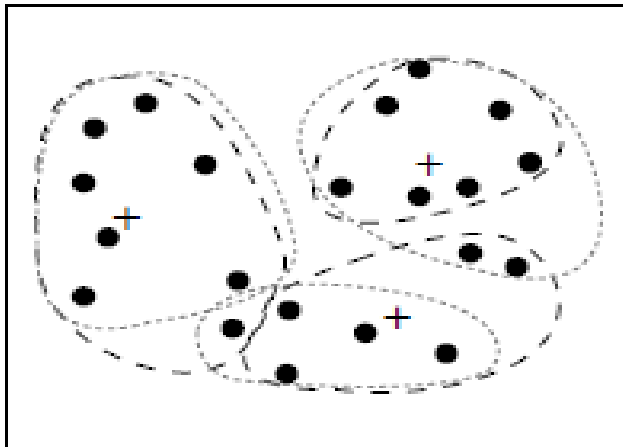
Method:

- 1) arbitrarily choose k objects from D as the initial cluster centers;
- 2) **repeat**
- 3) (re)assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster;
- 4) update the cluster means, that is, calculate the mean value of the objects for each cluster;
- 5) **until** no change;

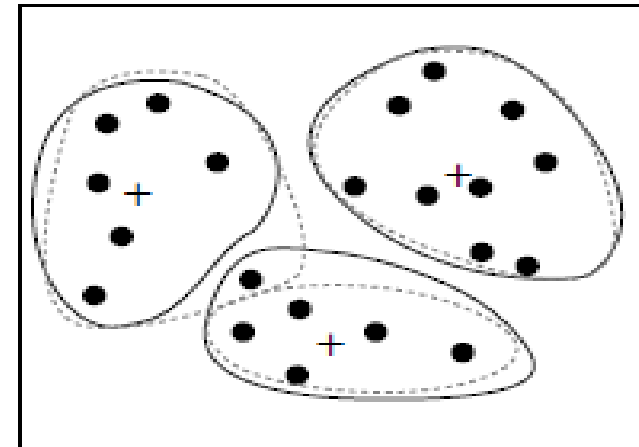
Algorithm: K-means



(a) Initial clustering



(b) Iterate



(c) Final clustering

Clustering of a set of objects using the k-means method; for (b) update cluster centers and reassign objects accordingly

K-means: Picking K

Heuristic: find the "elbow" of the within-sum-of squares (wss) plot as a function of K.

$$WSS = \sum_{i=1}^k \sum_{j=1}^{n_i} |x_{ij} - c_i|^2$$

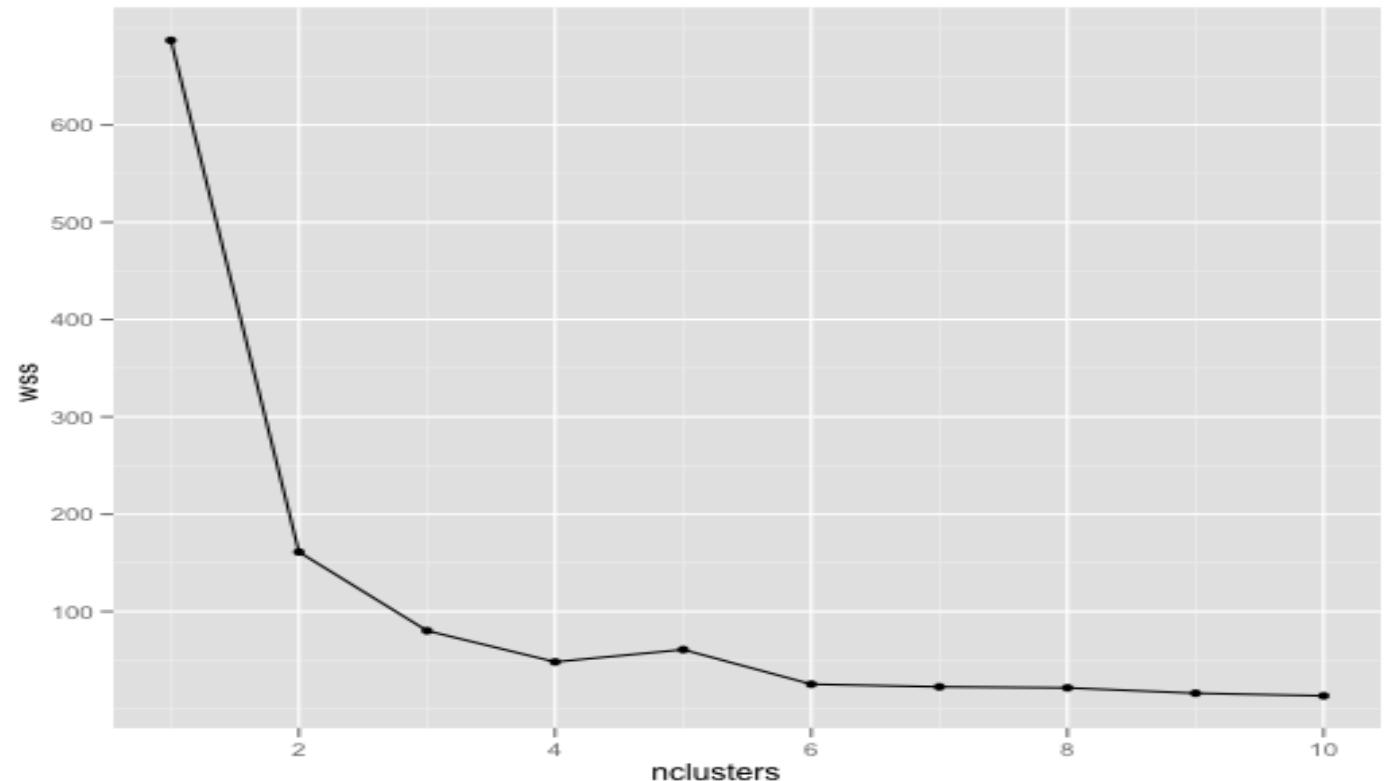
K: # of clusters

n_i : # points in i^{th} cluster

c_i : centroid of i^{th} cluster

x_{ij} : j^{th} point of i^{th} cluster

"Elbows" at $k=2,4,6$



K-Means Clustering - Reasons to Choose (+) and Cautions (-)

Reasons to Choose (+)	Cautions (-)
Easy to implement	Doesn't handle categorical variables
Easy to assign new data to existing clusters Which is the nearest cluster center?	Sensitive to initialization (first guess)
Concise output Coordinates the K cluster centers	Variables should all be measured on similar or compatible scales Not scale-invariant!
	K (the number of clusters) must be known or decided a priori Wrong guess: possibly poor results
	Tends to produce "round" equi-sized clusters. Not always desirable

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Summary

- A ***cluster*** is a collection of data objects that are similar to one another within the same cluster and are dissimilar to the objects in other clusters.
- The process of grouping a set of physical or abstract objects into classes of similar objects is called ***clustering***.
- Effective and scalable methods have been developed: partitioning method, hierarchical method, density-based method, etc.
- The k-means algorithm for partitioning, where each cluster's center is represented by the mean value of the objects in the cluster.

Thank you

